

Now click on Then That block and search for Adafruit on the search bar. Click on 'Send data to Adafruit IO' (see Fig. 10), select the feed name (ISS_LED), and type ON in the 'Data to save' field.

After configuration of Adafruit and IFTTT has been done, all that is left is to upload the code. To recap, we have created a toggle switch on Adafruit that can control your NodeMCU via Wi-Fi. And we have created an applet on IFTTT that toggles the Adafruit switch every time the ISS passes over the selected location.

The code

The heart of the ISS globe is the NodeMCU, which needs to be coded to work the way we want it to. Make sure you have the NodeMCU and Adafruit mqtt libraries installed on your Arduino IDE. The source code ISS_LED_SAMPLE.ino shown in Fig. 11 needs to be changed a bit for it to work.

Make following changes in the sample code:

- Replace SSID with your Wi-Fi (or mobile hotspot) name.
- Replace PASSWORD with your Wi-Fi (or mobile hotspot) password.
- Replace Adafruit ID with your Adafruit username.
- Replace AIO KEY with the active key you had noted after creating the Adafruit toggle switch.

If you named your Adafruit feed as ISS_LED you are done with editing of the code. But if you have used a different name, replace ISS_LED everywhere in the code with the name you have used.

This code keeps the LED turned off by default. When the ISS arrives, the LED starts blinking. In case you want the LED to remain on by default, you can replace `digitalWrite(led,LOW);` with `digitalWrite(led,HIGH);` in the first line of the void loop() function.

Now you can go ahead and upload the code to your NodeMCU. Please note, during testing, EFY Lab

YOUR ADAFRUIT IO KEY

Your Adafruit IO Key should be kept in a safe place and treated with the same care as your Adafruit username and password. People who have access to your Adafruit IO Key can view all of your data, create new feeds for your account, and manipulate your active feeds.

If you need to regenerate a new Adafruit IO Key, all of your existing programs and scripts will need to be manually changed to the new key.

Username:

Active Key: **REGENERATE KEY**

[Hide Code Samples](#)

Arduino

```
#define IO_USERNAME " "
#define IO_KEY " "

```

Fig. 8: Entering your Adafruit active key

IFTTT

Create your own

You've created 2 of 3 Applets

If This **Add**

Then That

Select trigger service

space

Search results

Space

Choose a location

923 Market St, San Francisco, CA 94103

Map showing location and nearby points of interest.

Fig. 9: Configuring If This block on IFTTT

Create your own

You've created 2 of 3 Applets

If ISS passes over a specific location

Then That **Add**

Select action service

adafruit

Search results

Adafruit

Select action

Adafruit

Send data to Adafruit IO

Complete action fields

Send data to Adafruit IO

This Action will send data to a feed in your Adafruit IO account.

Feed name: ISS_LED

The name of the feed to save data to.

Data to save: ON

The data to be saved to your feed.

Continue

Fig. 10: Configuring the Then That block on IFTTT